

2025 Integrated Work Study Programme (IWSP) at OMRON as a Robotics Engineering Intern

ABOUT OMRON

OMRON Singapore (OEPSG) is an industrial automation partner that creates, sells and services fully integrated automation solutions that include sensing, control, safety, vision, motion, robotics and more.



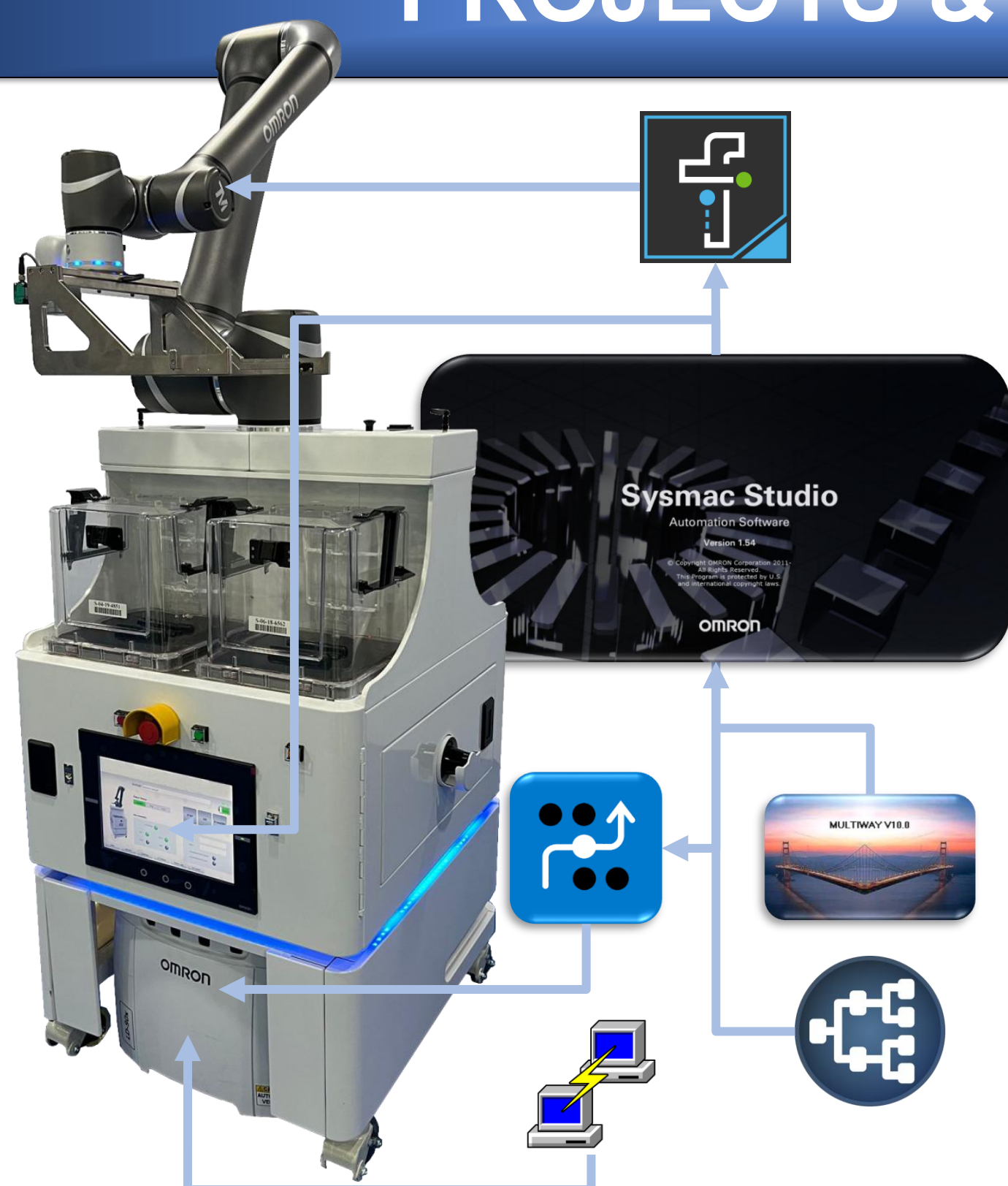
OMRON SG SG60 Photoshoot

JOB SCOPE

- Assist project team in performing duties:
- Supporting engineers at customer site
 - Monitoring of deployed robotics solutions
 - Debugging issues observed/ raised by customers
 - Performing simulations and in-lab validation

INTERNSHIP JOURNEY: A LEARNING EXPERIENCE

PROJECTS & TASKS



Mobile Manipulator (MoMa) robot introduces **Industrial Automation** of wafer cassette handling, replacing a previously labour-intensive process.

It orchestrates round-the-clock operations across 22 machines sitewide, increasing throughput by 10%

MoMa also frees capacity, enabling workers to be redeployed and upskilled to perform at value-added roles with skillsets that advance their careers.



Created a generic Application Programming Interface (API) script for CPP Inference of Open Neural Network Exchange (ONNX) formats on OMRON controllers using ONNX Runtime (ORT) & OpenVINO for optimization of inference speeds.



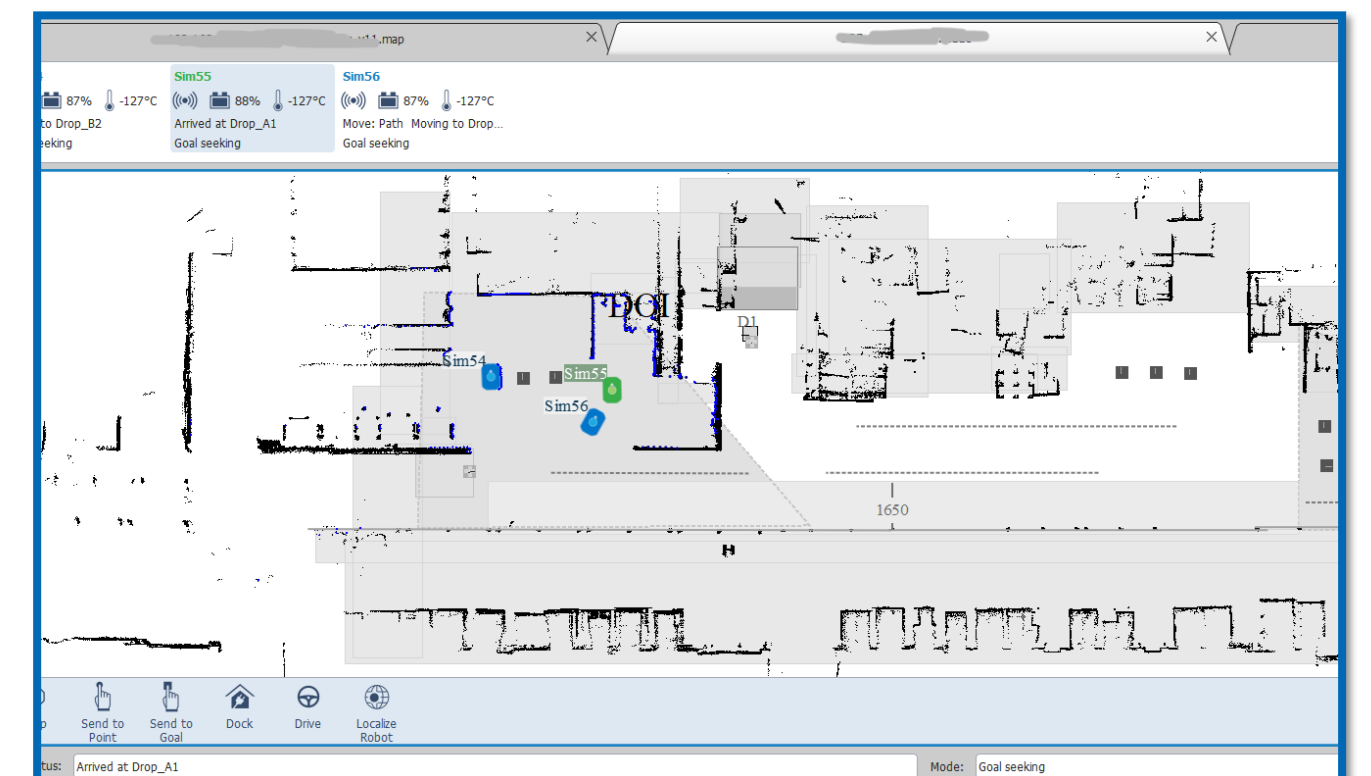
Wrote a script for 2D image processing and display of OMRON SENTECH Liquid Lens Industrial Cameras using OpenCV and StApiPy.



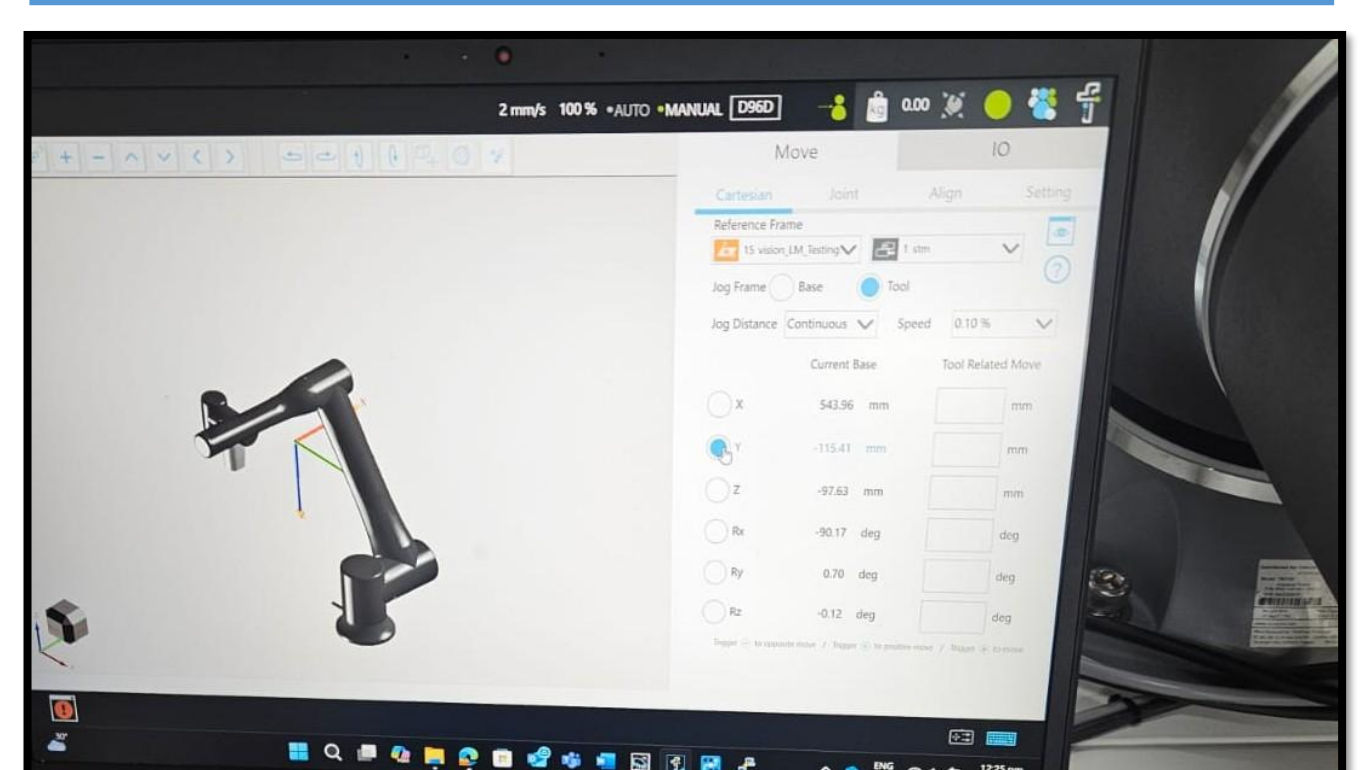
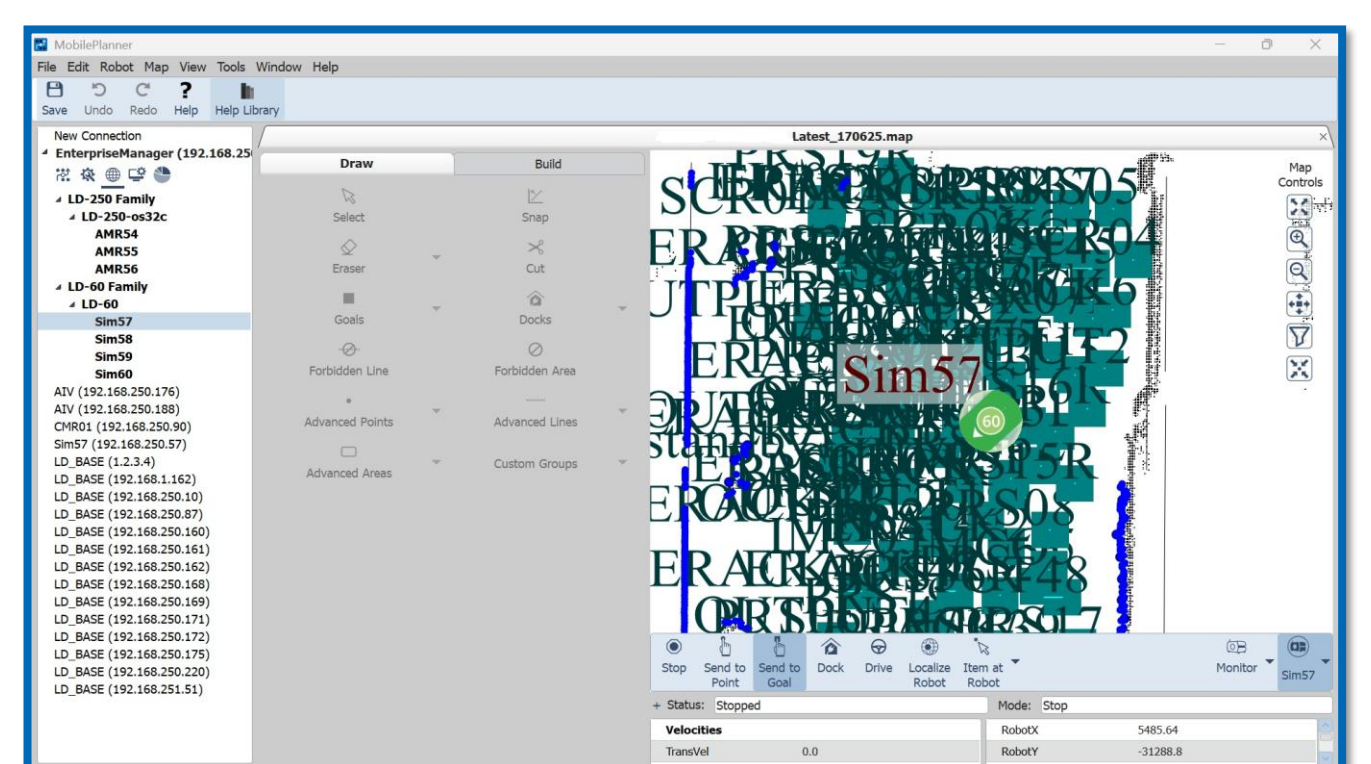
OMRON Team Bonding Day



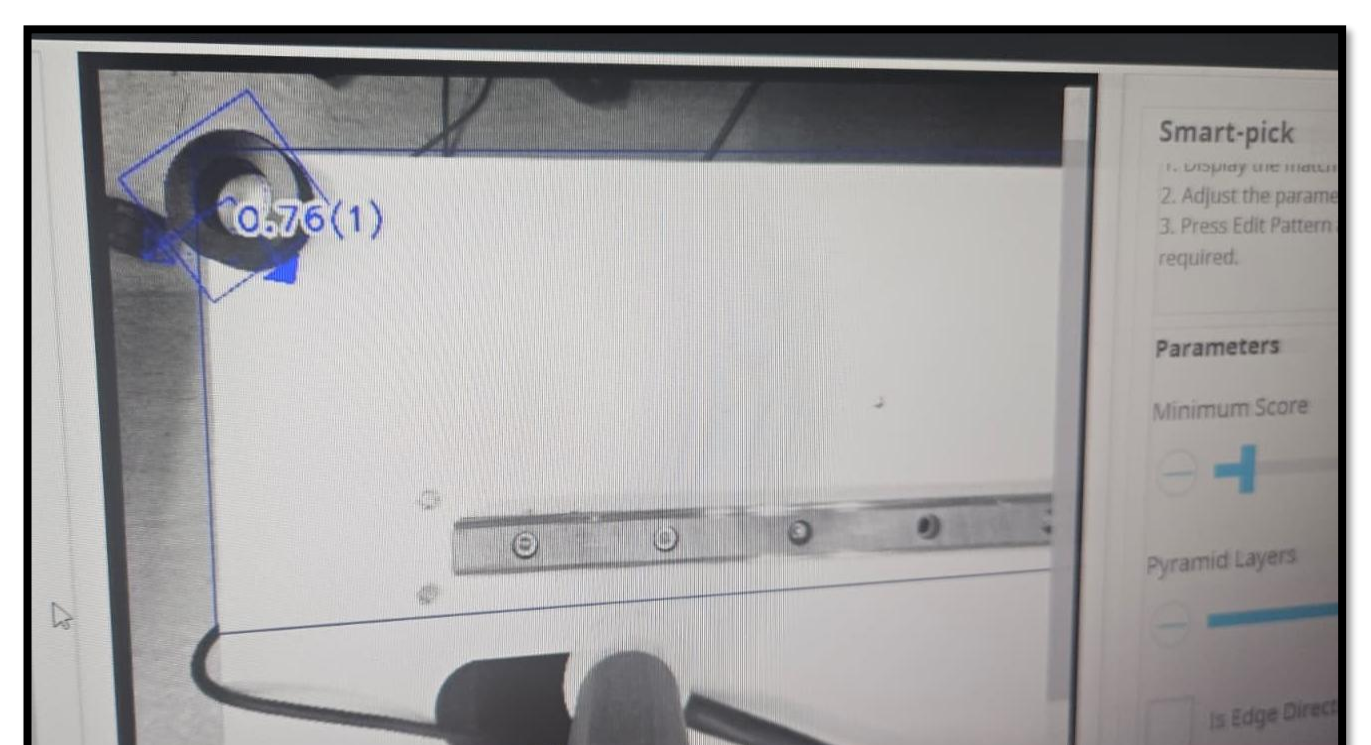
OMRON International Sushi Day



Created Digital Twin simulation scenarios in **Mobile Planner software** using Virtual Fleet Manager (VFM) for path-planning, traffic management and in-lab validation prior to site deployment.



Built Pick-and-Place routines in **TMFlow software** on OMRON TM Robotic arm and delivered stakeholder Proof-Of-Concept (POC) demonstration videos for picking circular, hollow-centre objects.



LESSONS LEARNT

Working alongside engineers on various projects & tasks gave me exposure to *professional industry experience* contrasting beyond being an *academic* in a classroom. One key lessons learnt is that the **learning never stops**.

The process of **relearning** to apply what is taught in a new, unfamiliar environment may seem daunting at first, but adopting an open, critical mind, I learn to translate theory into application and **keep moving forward**.